



USAII GLOBAL AI HACKATHON 2026

SentinelX

AI-assisted threat triage and contextual exploitation analysis for NGOs and intervention organizations

Graduate Track

Masters & PhD

Challenge Brief 5

Build AI Systems That Protect People

Direction A

Safe Passage — AI Against Digital Exploitation

Build window: June 14 – 21, 2026 | Fully virtual, global | Team of 5

Warning signs exist long before intervention happens

Organizations protecting vulnerable people must make high-stakes decisions with incomplete, fragmented, and sensitive data — often too late to prevent harm. Digital exploitation, coercion, and grooming are increasingly hard to see until damage is already done.

Operational reality our system must respect

- Data is fragmented, delayed, and incomplete
- Signals are weak, ambiguous, or noisy
- Organizations face limited staff, capacity, funding
- Privacy, trust, and legal boundaries restrict sharing
- Action stays constrained even when risk is detected



Reactive, not preventive

Detection typically happens after escalation, because risk signals sit fragmented across disconnected systems and reports.



False positives cause real harm

Misclassifying a vulnerable person's situation can trigger panic, damage trust, or provoke an inappropriate organizational response.

Our users: NGO analysts and investigators at protection-focused organizations — not surveillance of individuals.

Built for the people who show up first

As warning signs of stalking, coercion, and exploitation increasingly move online, organizations struggle to process fragmented reports fast enough to respond. SentinelX is designed around one principle: AI should help analysts see patterns earlier — humans decide what happens next.



NGO analysts

Review intake reports, validate AI-surfaced insights, decide on escalation



Intervention organizations

Coordinate response across stalking, coercion, and exploitation cases



Vulnerable individuals

Submit structured reports safely, without heavy media uploads or exposure



Protection-focused teams

Need explainable, auditable reasoning — not opaque automated verdicts

Three narrowly-scoped flows, not one black box

Rather than one autonomous pipeline, SentinelX separates concerns by risk level — each flow has its own human control point.



SOS

No AI — instant alert

3-second long-press sends an immediate alert to emergency contacts and the NGO dashboard simultaneously. Kept deliberately simple for true emergencies.



Detect Threat

On-device AI — opt-in only

User-toggled. Audio is transcribed locally and discarded. A Bi-LSTM classifies text as threat / non-threat on-device. It only flags — never escalates.



Safe Chat + Evidence

Analyst-gated AI

Guided 4-step intake plus optional evidence upload. Stored as 'Pending Analysis' until an analyst clicks [Analyze Report] — Gemini never runs without that click.

Why AI — and why two separate models

Threat, coercion, and grooming patterns depend on repetition, timing, and narrative framing — signals a keyword filter cannot reliably catch. We use two independent, narrowly-scoped AI components instead of one general pipeline.



Bi-LSTM — on-device classifier

Trained on 100,000+ labeled statements (Kaggle threat/non-threat corpus) — 70/20/10 split, **96% test accuracy**.

- Powers Detect Threat — runs entirely on the user's phone
- Classifies locally-transcribed text as threat / non-threat
- Selected for sequential language understanding, not keyword matching
- Never sees, stores, or transmits raw audio



Gemini API — evidence analysis

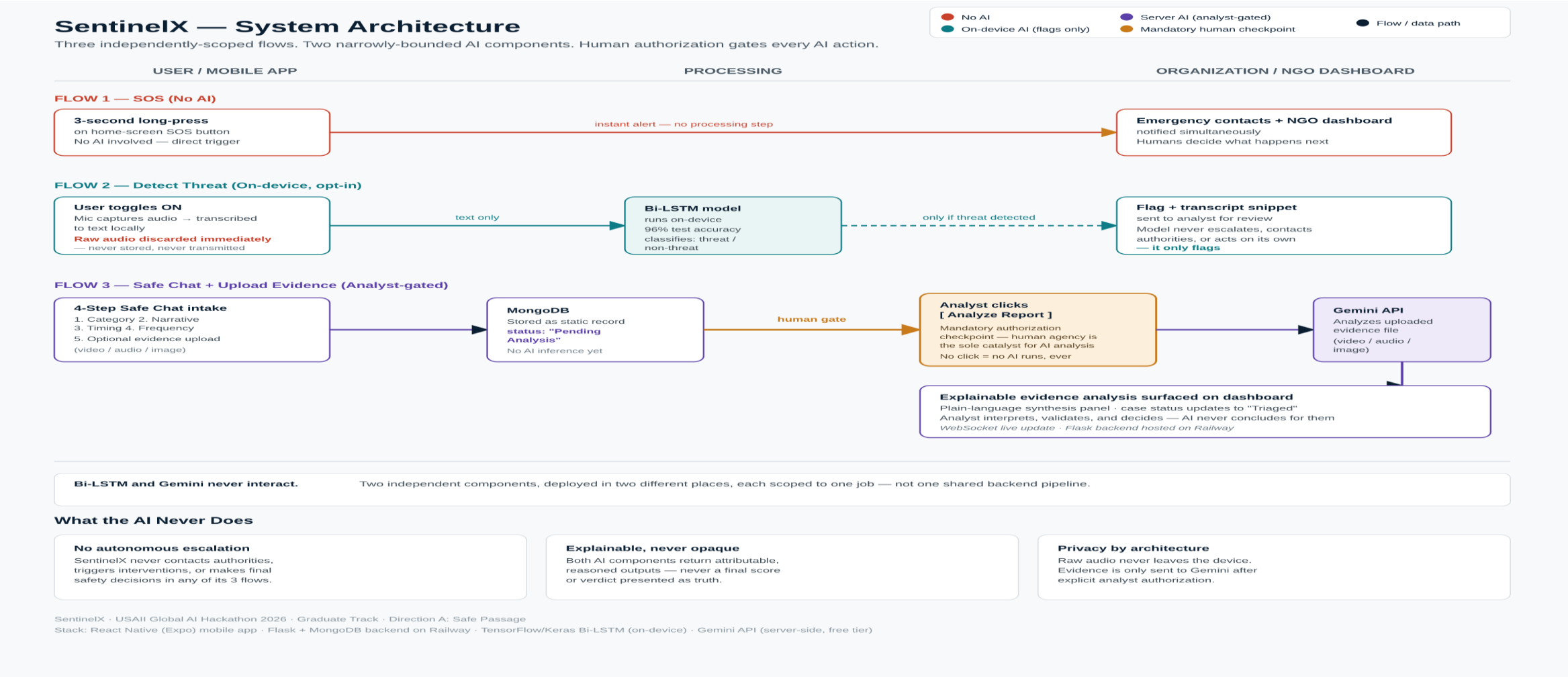
Runs server-side, **only after an analyst clicks [Analyze Report]**.

- Analyzes uploaded video, audio, or image evidence
- Selected because it reasons over unstructured media —
- rules engines can't interpret video or audio context
- Returns plain-language, analyst-readable justification
- Inference-only: not trained or fine-tuned on user data

The Bi-LSTM and Gemini never interact — two completely separate features, separate failure domains.

System architecture

Signal → Insight → Decision → Action → Outcome, with a mandatory human gate before any AI analysis



The decision SentinelX intentionally does not make

SentinelX never autonomously determines whether a case should be escalated — to NGOs, law enforcement, or emergency responders — in any of its three flows.

1

SOS

No AI decision exists. The alert goes directly to humans — what happens next is entirely up to emergency contacts and the analyst.

2

Detect Threat

The user's own opt-in toggle is the control point. The model only flags a transcript snippet for analyst review — it never acts.

3

Safe Chat + Evidence

The analyst's explicit [Analyze Report] click is the control point. Gemini never analyzes evidence until a human asks it to.

If our AI gets it wrong — what happens to the user?

A weak answer says *"we tested it and it works fine."* A strong answer names the specific failure mode, who gets harmed, and the design choice that reduces it. We answer with the second.

Failure mode: automation bias

Who gets harmed: an analyst over-trusts an AI flag or evidence summary as definitive truth during a sensitive case, leading to a false escalation or a missed one.

Design choice: both AI components produce explainable, attributable outputs — never opaque final classifications. Neither can act on its own. The Bi-LSTM only flags; Gemini only describes. The analyst remains the final decision-maker, every time.

Failure mode: surveillance creep via Detect Threat

Who gets harmed: a vulnerable user, if continuous audio capture is poorly scoped — the exact surveillance tool this challenge warns against.

Design choice: strict data minimization. Audio is transcribed on-device and discarded immediately — never stored or uploaded. Only a threat/non-threat label and transcript snippet leave the device, and only when a threat is detected. Detect Threat is opt-in, never silent or default-on.

What SentinelX explicitly does not do



Scope & non-goals

- Does not autonomously contact authorities or trigger interventions
- Does not perform passive or continuous background monitoring
- Does not store or transmit raw audio, in any flow
- Does not train any model on real survivor or victim case data
- Does not present AI output as a final verdict or risk score



Safeguards built into the system

- Mandatory human-triggered evidence analysis — analyst clicks [Analyze Report]
- No autonomous intervention escalation in any of the three flows
- Structured, consent-based intake — never passive background monitoring
- Explainable, attributable outputs instead of opaque blended scores
- Transparent analyst review interfaces with full audit context

What changes for the analyst because SentinelX exists

BEFORE

- Reports arrive fragmented across channels, hard to compare
- Analysts manually re-read evidence with no structured support
- Risk signals identified only after escalation already happened
- No consistent, auditable reasoning trail for case decisions

AFTER

- Structured 4-step intake produces consistent, comparable signals
- Gemini surfaces a plain-language summary of uploaded evidence
- Threat-language flags arrive in real time for analyst triage
- Every case carries an explainable, attributable reasoning trail

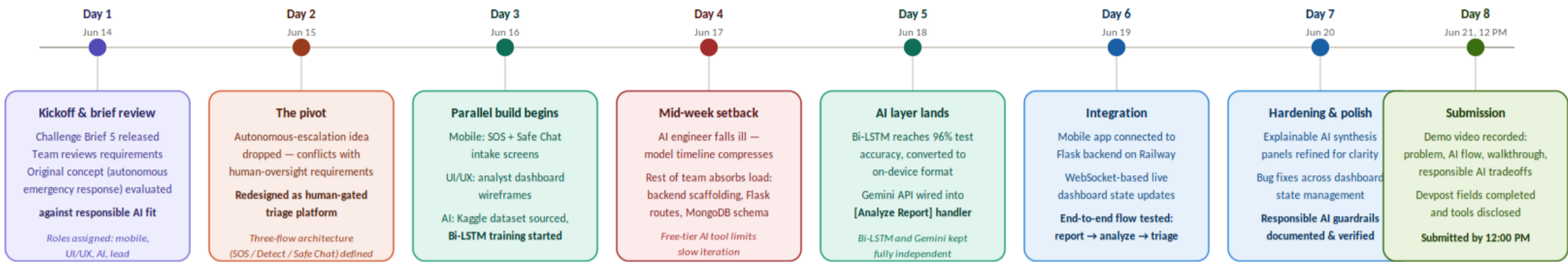
HOW WE BUILT IT

Build roadmap

June 14, 12:00 PM → June 21, 12:00 PM — pivot, parallel build, and a mid-week setback we built around

SentinelX build roadmap

USAII Global AI Hackathon 2026 — Graduate Track, Challenge Brief 5 — June 14, 12:00 PM to June 21, 12:00 PM



Parallel workstreams across the build window

Mobile (React Native / Expo)
SOS long-press · Safe Chat 4-step wizard · evidence upload · on-device Bi-LSTM bundling

Dashboard / UI-UX
Analyst case queue · [Analyze Report] gate · explainable synthesis panels · alert feeds

AI / ML (Bi-LSTM + Gemini)
100K-statement training corpus · TensorFlow/Keras training · on-device conversion · Gemini integration

Backend (Flask / MongoDB / Railway)
Report ingestion · case state machine · WebSocket sync · Gemini routing on analyst trigger

Responsible AI thread (continuous, all 8 days)
Human-gate design · automation-bias review · data minimization decisions · scope/non-goals documentation

Who built it, and what it's built on



Team lead

Architecture pivot, responsible AI design, coordination



Mobile developer

React Native (Expo) app — SOS, Safe Chat, on-device Bi-LSTM



UI/UX designer

NGO analyst dashboard, triage workflows, explainable AI views



AI Lead

Bi-LSTM training pipeline, Gemini integration, contextual NLP

Stack — built entirely on free-tier tools

AI & models

- Bi-LSTM (TensorFlow / Keras)
- Gemini API (free tier)

Backend

- Python + Flask
- MongoDB
- Railway (free tier hosting)

Frontend

- React Native (Expo)
- WebSocket live dashboard updates

Dev assistance

- ChatGPT, GitHub Copilot,
- Gemini, Claude, Lovable
- (all free-tier / student access)

No paid enterprise tools were used.

Toward responsible AI safety infrastructure at scale

- Improve contextual threat synthesis accuracy and multilingual analysis
- Strengthen explainable AI reporting for analyst collaboration
- Explore cross-case escalation trend analysis with privacy boundaries
- Build stronger privacy-preserving case management infrastructure



AI should assist human judgment in high-stakes safety environments — not replace it.

SentinelX

TEAM

Who built SentinelX

A five-person team, coordinating across the one-week build window.



FREDRICK JOHN
Team Lead

Architecture, responsible AI
design, coordination

IORHEN TOMNA
AI/ML Lead

Bi-LSTM training pipeline,
model architecture

ADEYANJU ADEBAYO
Asst. AI/ML Lead

Gemini integration, evaluation,
contextual NLP

AHMAN SOLOMON
Mobile Developer

React Native (Expo) app —
SOS, Safe Chat, Detect

OGUNYEMI HABEEB
UI/UX Designer

NGO analyst dashboard, triage
workflows, XAI views